

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools required

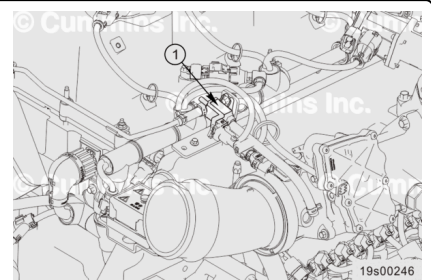
Additional Service Items

- No additional service items required.

General Information

The turbocharger wastegate position and boost pressure are controlled by the wastegate control valve (1).

In default position, the wastegate control valve directs full boost pressure to the wastegate actuator.



Preparatory Steps

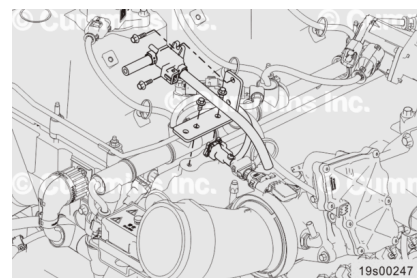
- Disconnect the turbocharger actuator air line. Refer to Procedure 010-118 in Section 10. ([/qs3/pubsys2/xml/en/procedures/1016/1016-010-118.html](https://qs3.pubsys2/xml/en/procedures/1016/1016-010-118.html))

Remove

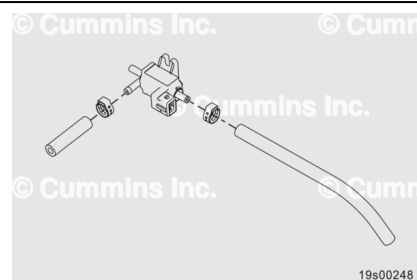
Disconnect the engine harness connector from the wastegate control valve.

Remove the two capscrews connecting wastegate control valve to the bracket.

Remove the two capscrews connecting wastegate control valve bracket to the intake manifold.



Remove and discard the hose clamp connecting wastegate control valve to hose if required.

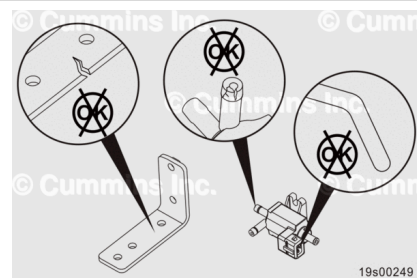


Inspect for Reuse

Inspect the wastegate control valve for bent or broken pins or damaged connections.

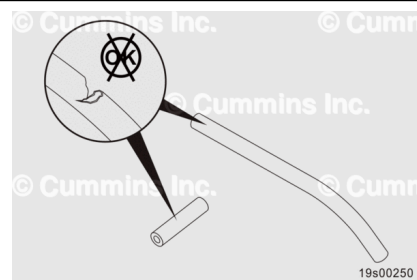
Inspect the wastegate control valve bracket for cracks or other damage.

Replace the wastegate control valve or bracket if damage is found.



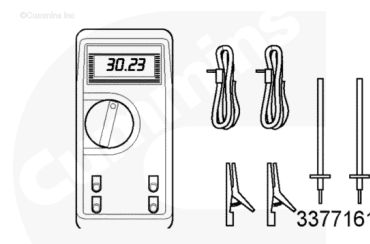
Inspect all hoses for cracks, cuts, or other damage that can cause leaks.

Replace hose if damage is found.



Test

Measure the resistance between the signal and return pins at the wastegate control valve connector.



Wastegate Control Valve Body Temperature		Resistance (Ohms)	
°C	°F	MIN	MAX
-40	-40	65.5	72.4
-35	-31	67.2	74.2
-30	-22	68.8	76.1
-25	-13	70.5	77.9
-20	-4	72.2	79.8
-15	5	73.8	81.6
-10	14	75.5	83.4
-5	23	77.2	85.3
0	32	78.8	87.1
5	41	80.5	89.0
10	50	82.2	90.8
15	59	83.8	92.7
20	68	85.5	94.5
25	77	87.2	96.3
30	86	88.8	98.2
35	95	90.5	100.0
40	104	92.2	101.9
45	113	93.8	103.7
50	122	95.5	105.6
55	131	97.2	107.4
60	140	98.8	109.2
65	149	100.5	111.1
70	158	102.2	112.9
75	167	103.8	114.8
80	176	105.5	116.6

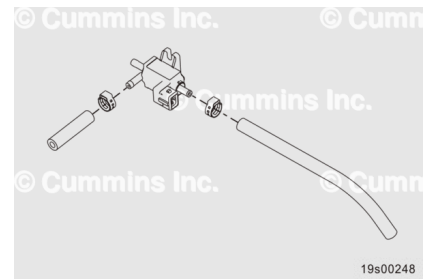
Measure the resistance between the signal pin and wastegate control valve body.

Minimum resistance: 100k ohms.

If the resistance does **not** meet the specifications, replace the wastegate control valve.

Install

Install new hose clamp connecting wastegate control valve to hose if removed.



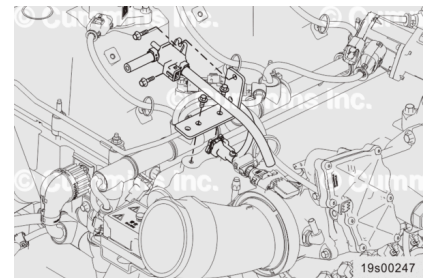
Install the two capscrews connecting wastegate control valve bracket to the intake manifold.

Torque Value: 5 n•m [44 in-lb]

Tighten the two capscrews connecting wastegate control valve to the bracket.

Torque Value: 5 n•m [44 in-lb]

Connect the engine harness connector to the valve.



Finishing Steps

- Connect the turbocharger actuator air line. Refer to Procedure 010-118 in Section 10. (</qs3/pubsys2/xml/en/procedures/1016/1016-010-118.html>)